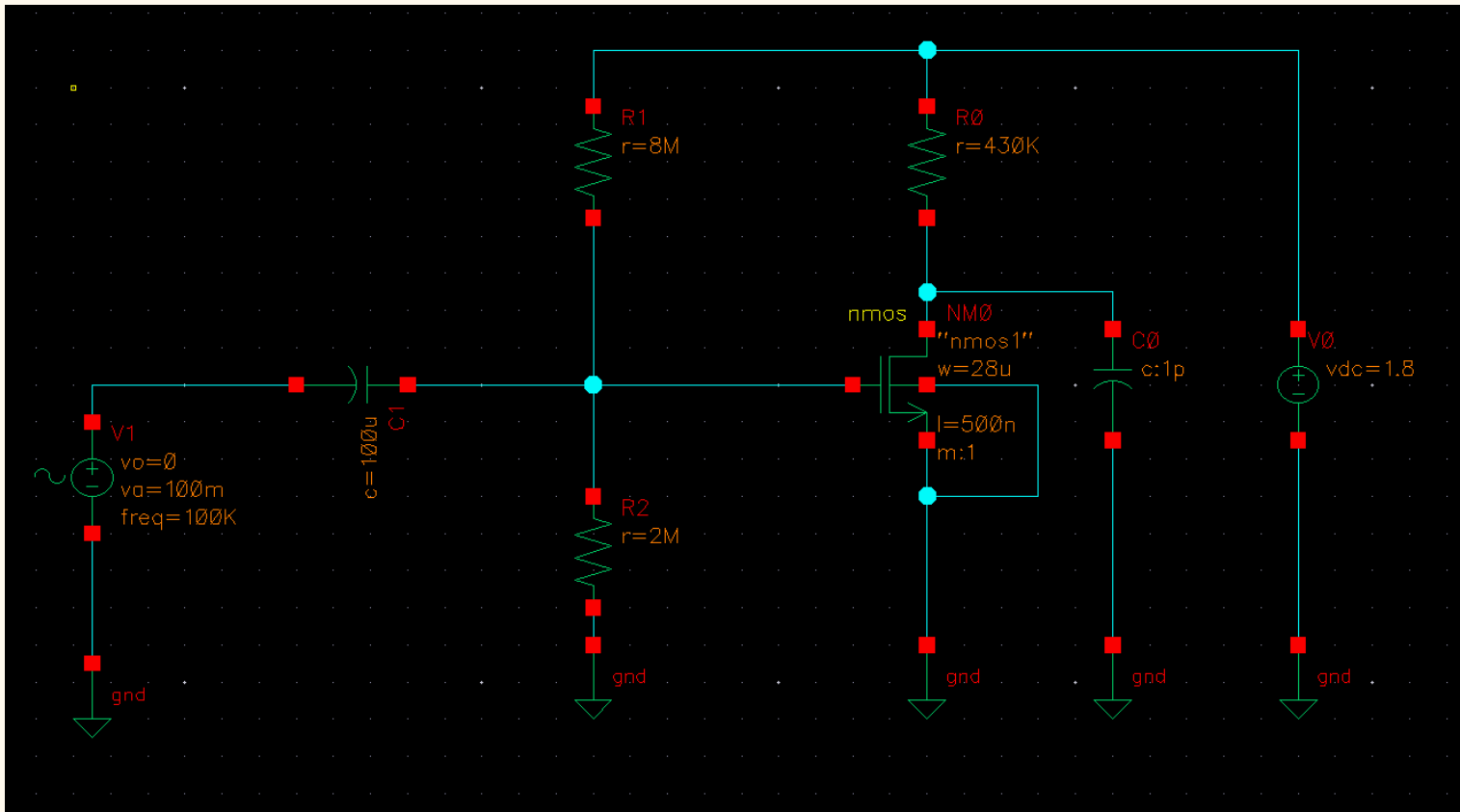
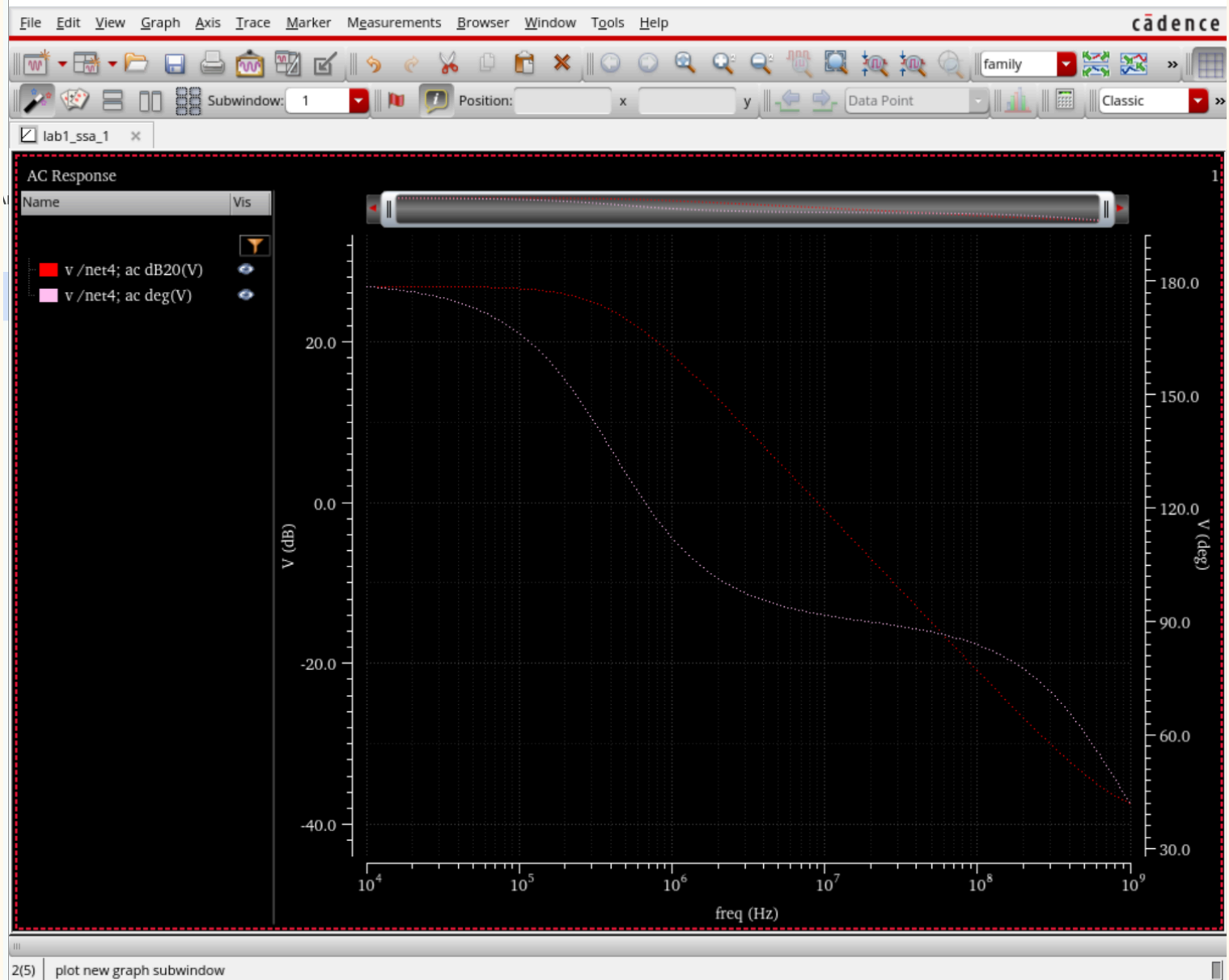


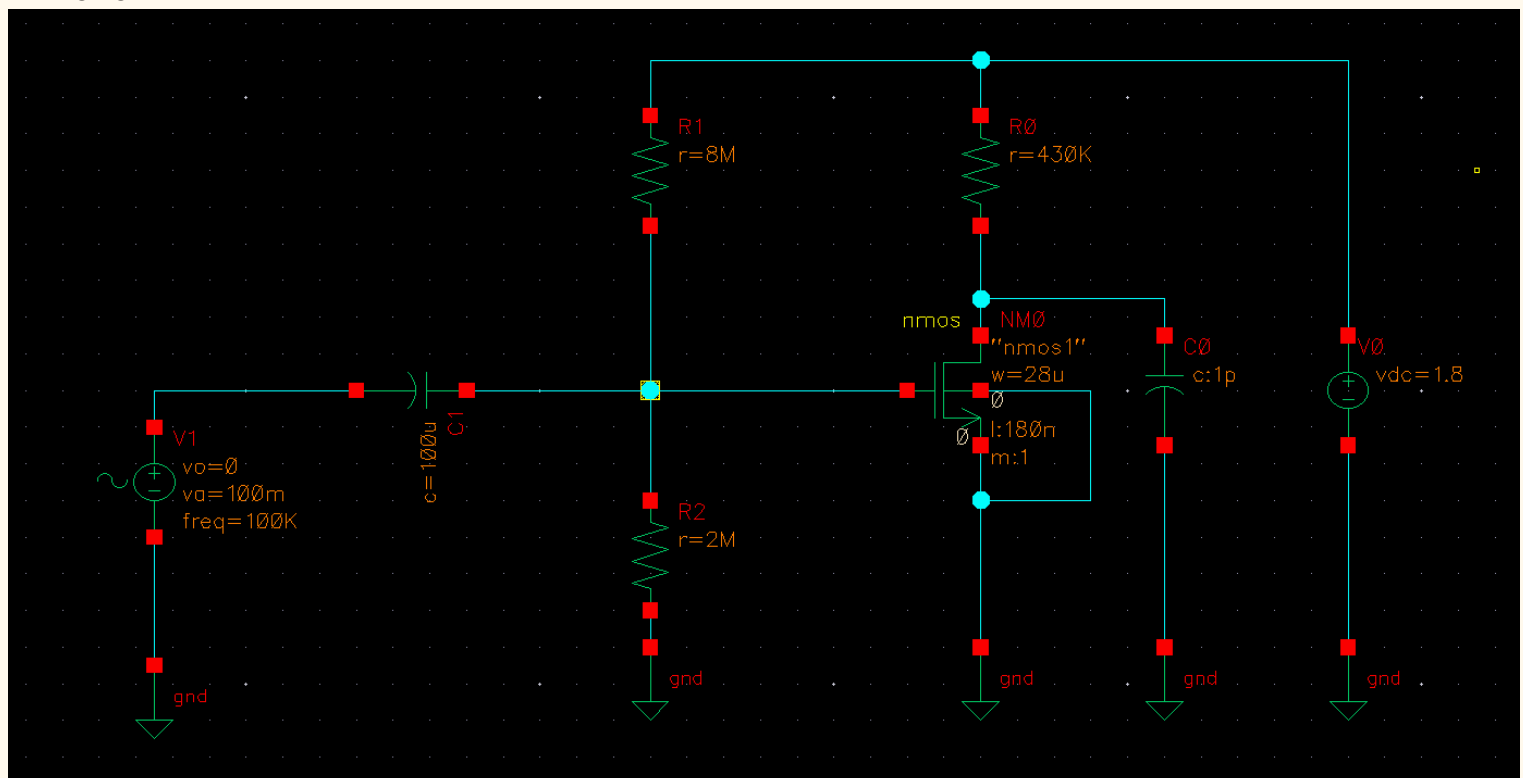
Part 1 SSA

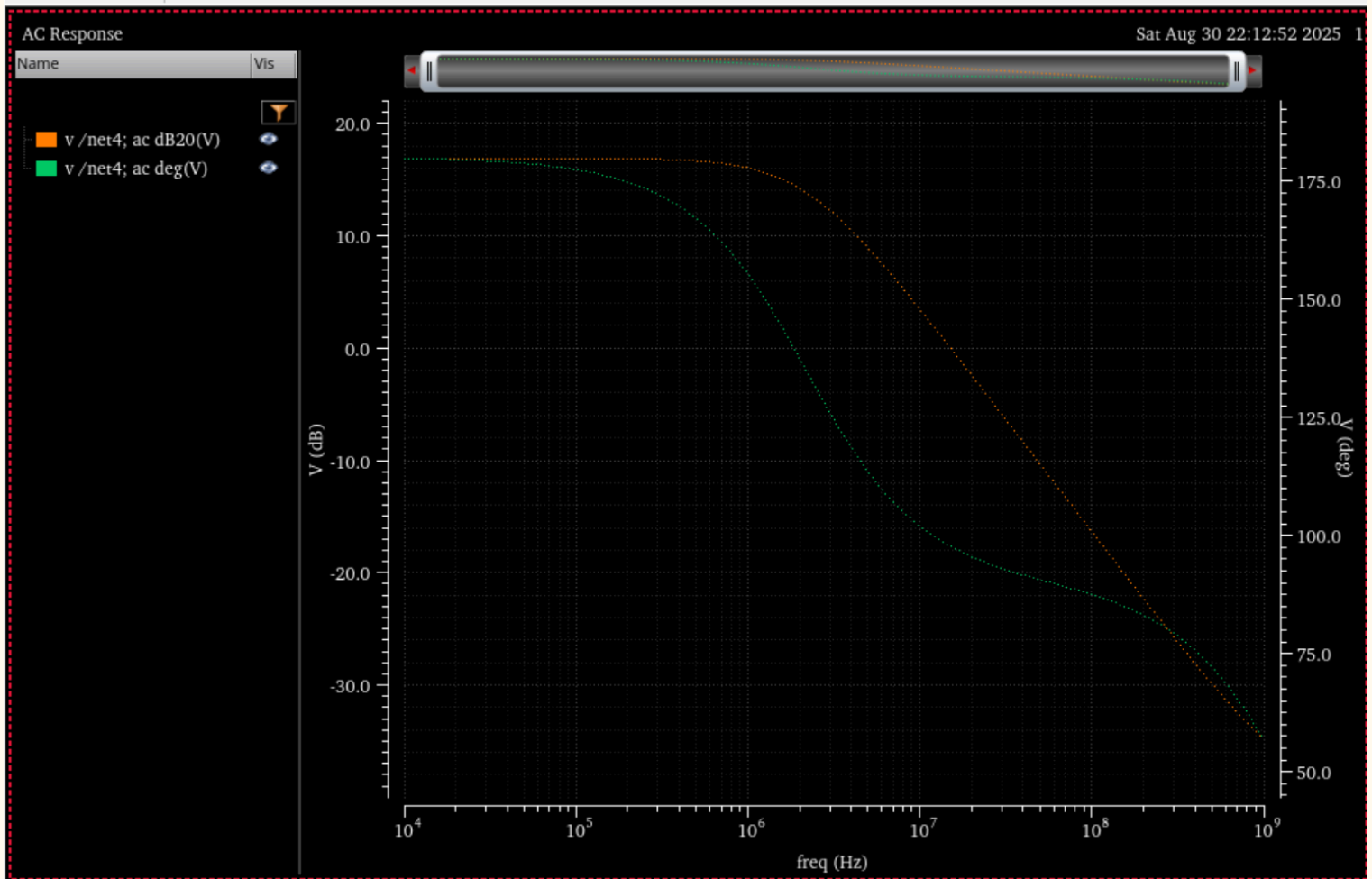
AC





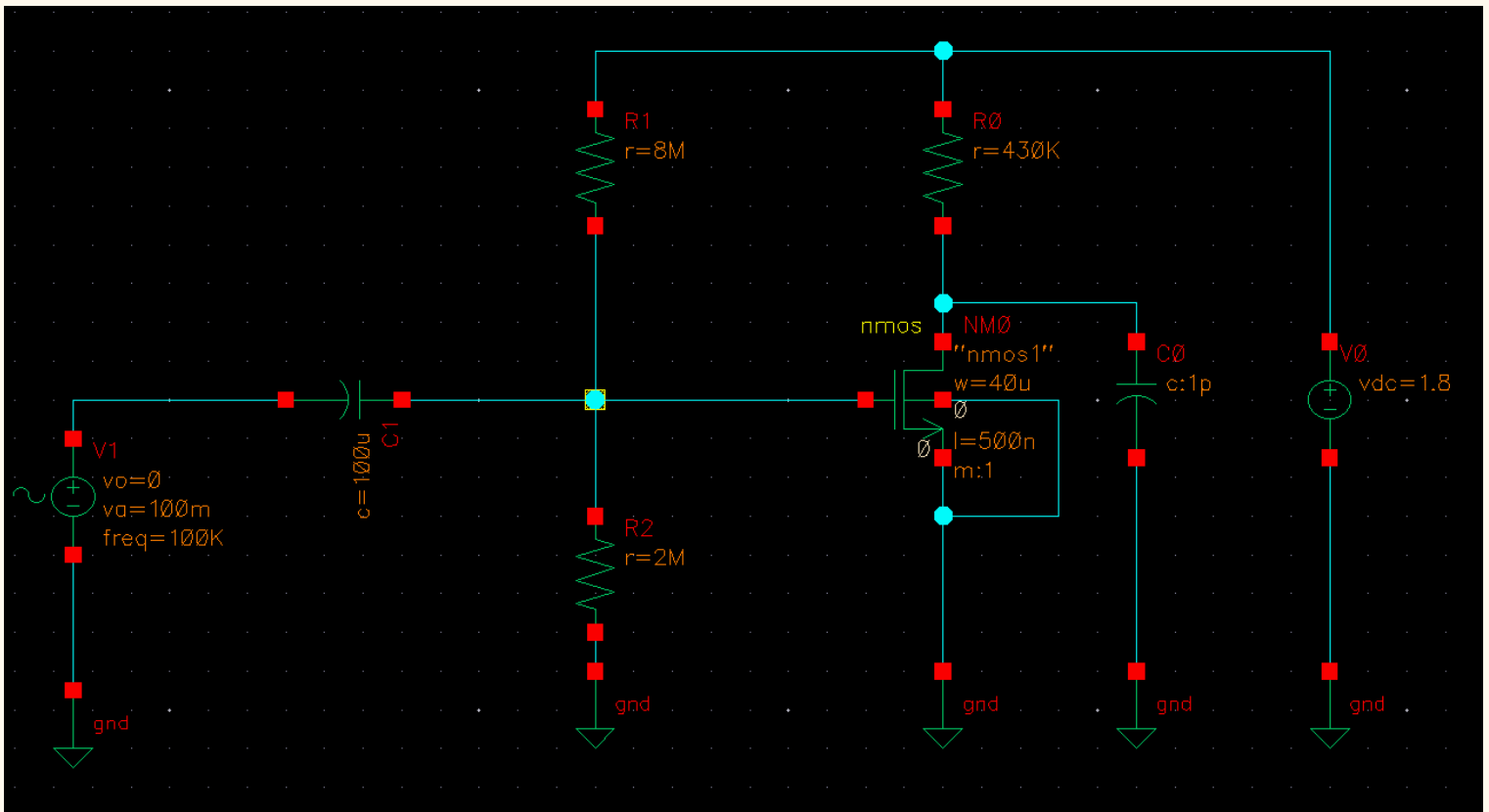
Changing L from 500nm to 180nm

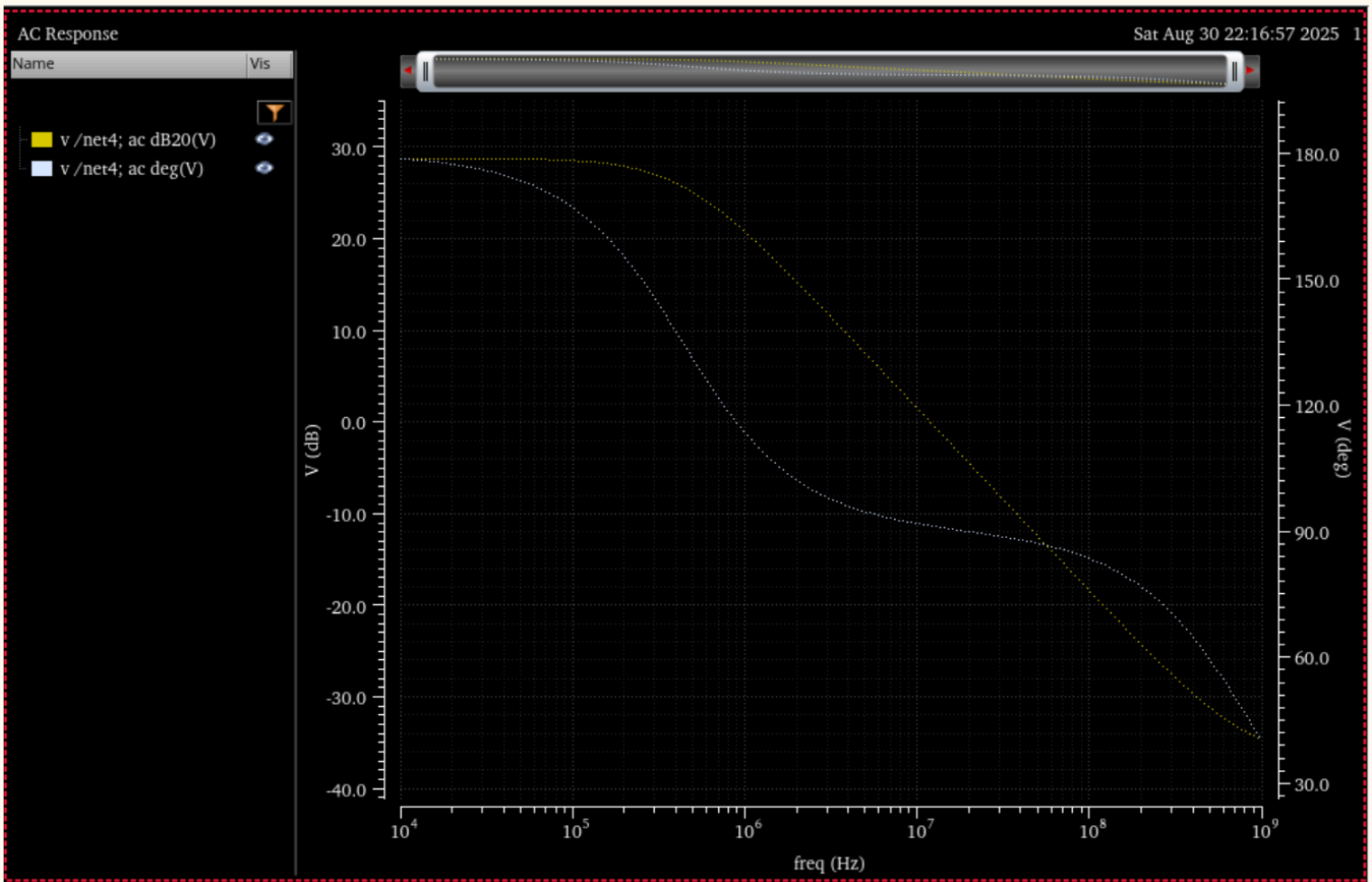




- BW increased
 - smaller L smaller C

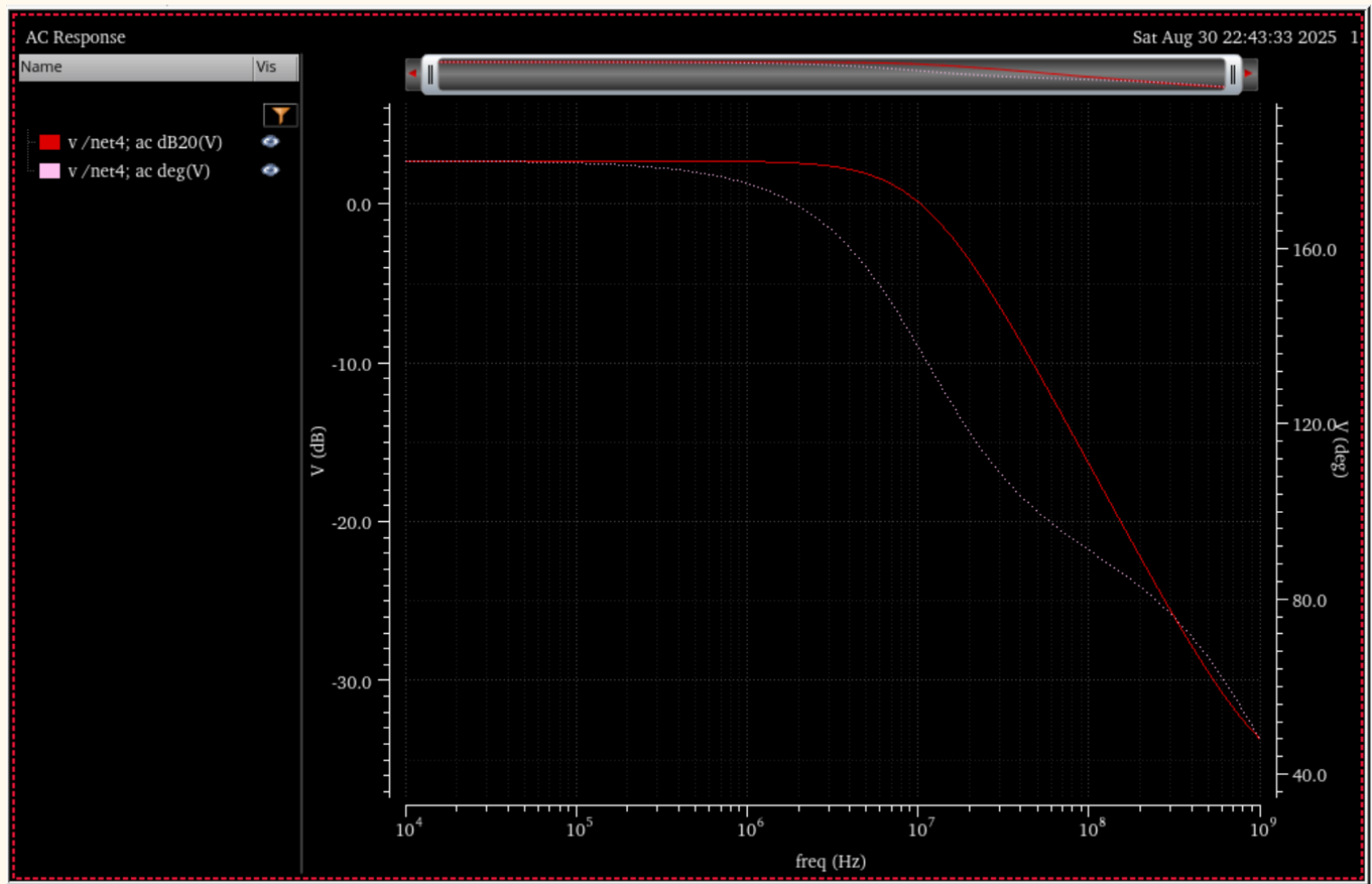
Changing W from 28um to 40um





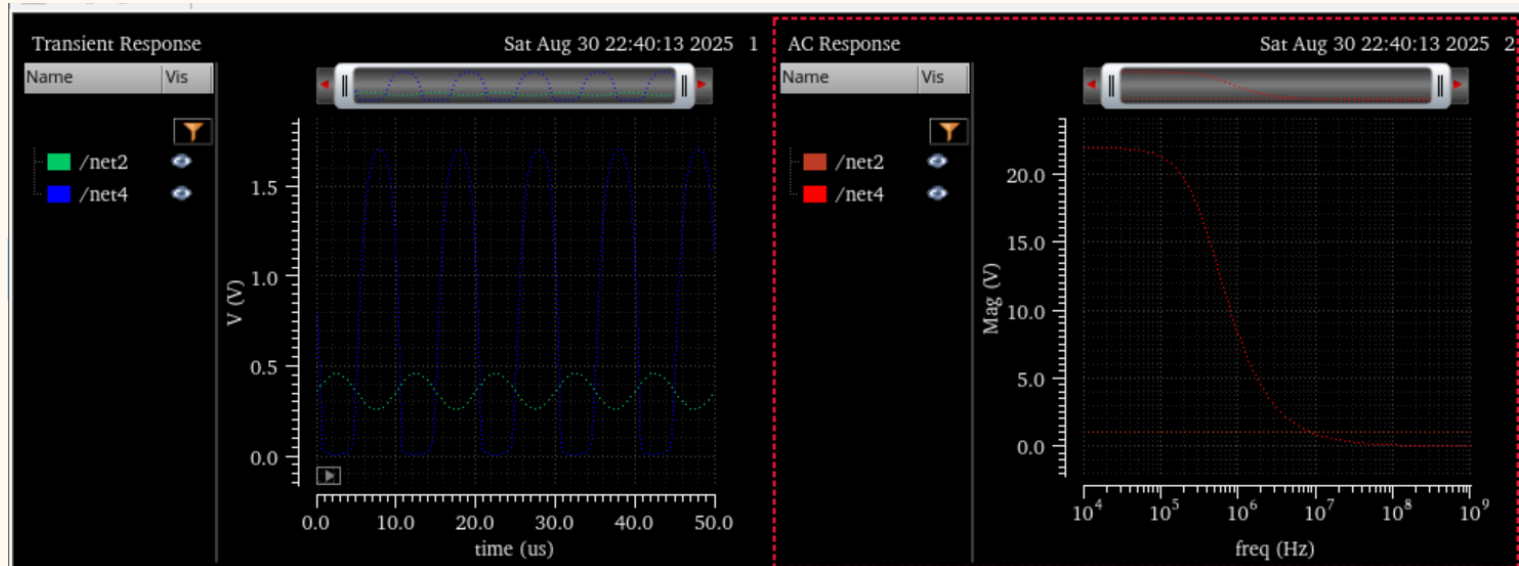
- gain increased
- BW like same same?

Changing W from 28um to 40um and L from 500nm to 180nm

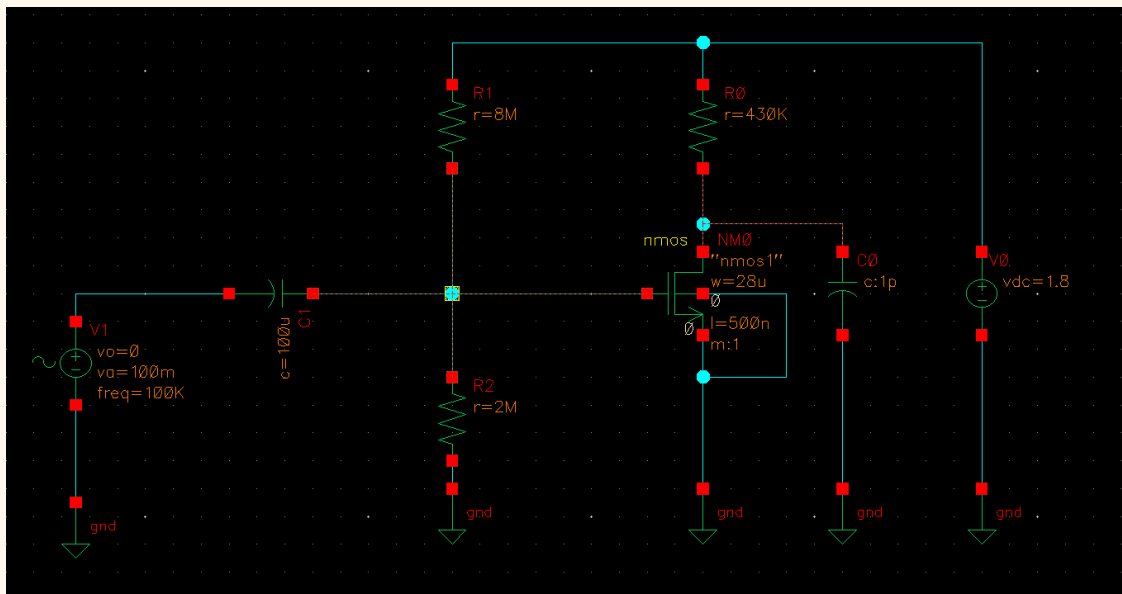


- much lower gain but wider bandwidth

Transient



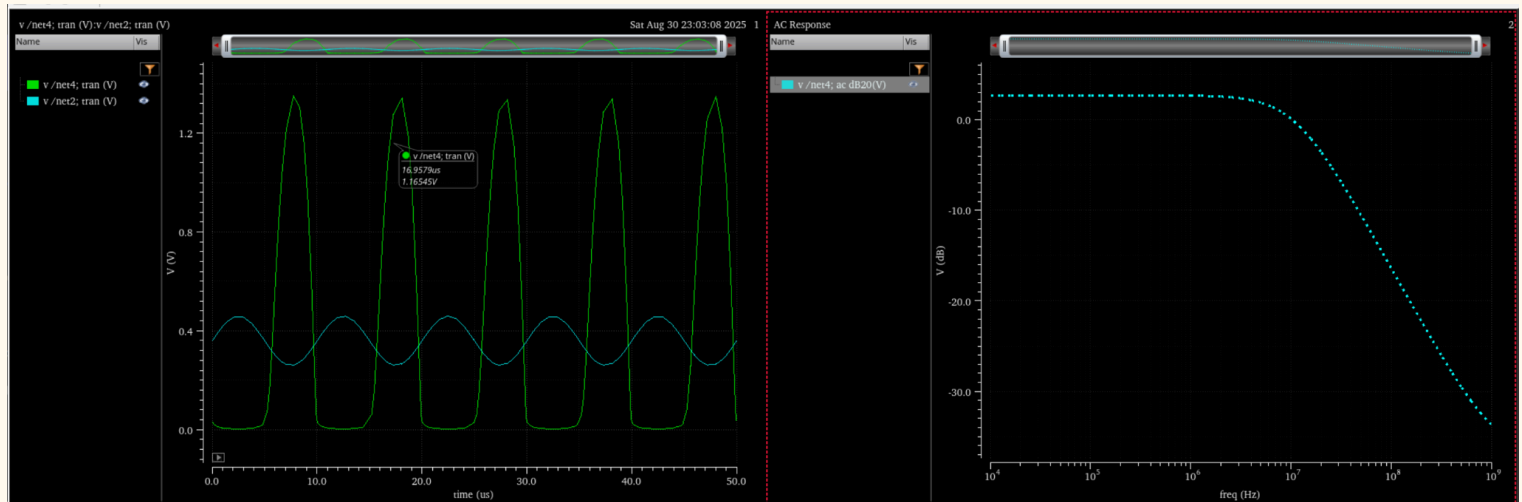
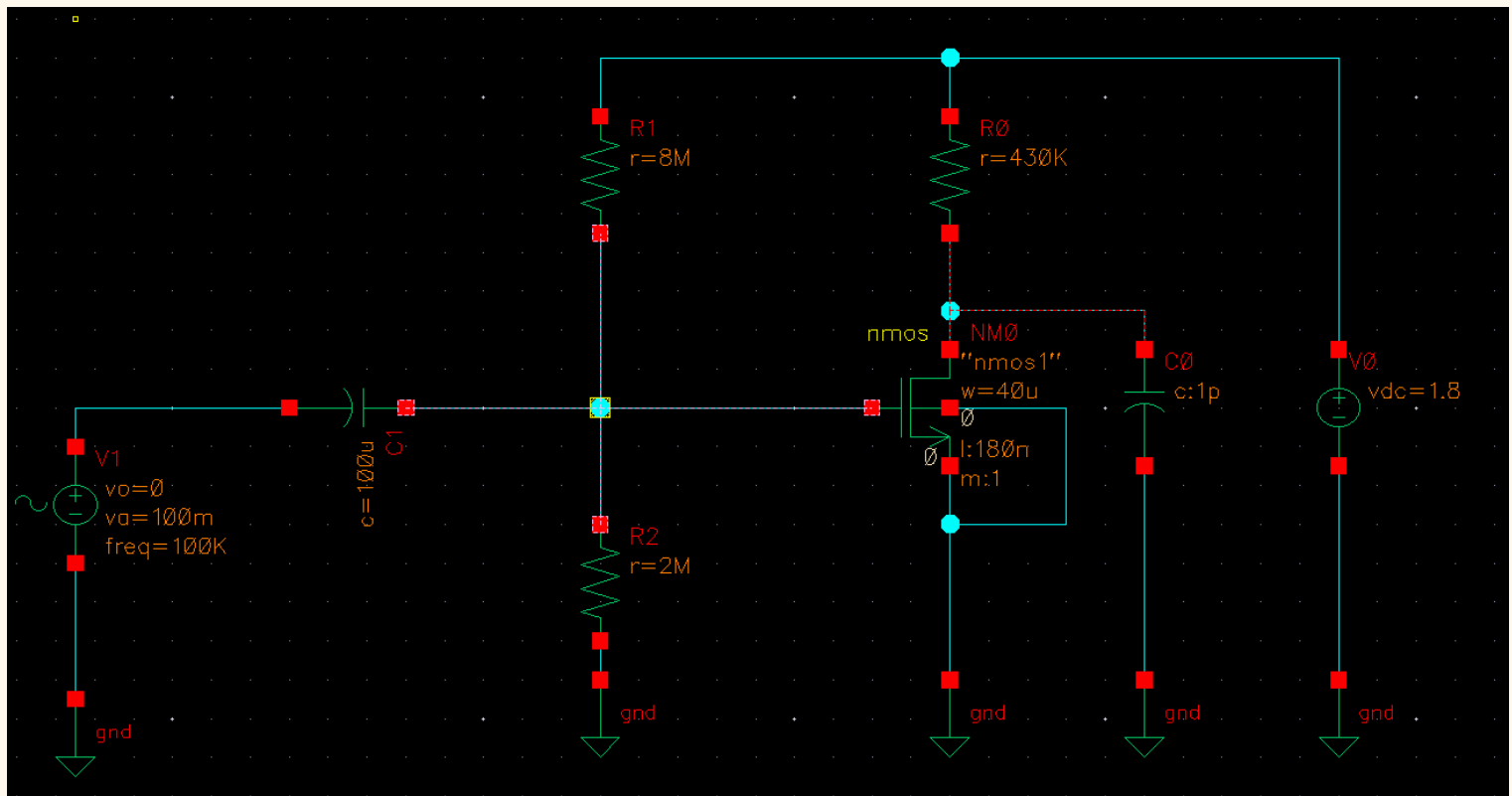
- setup the save results to auto plot
- wtf eh its clipping
 - with the original design



○

https://ocw.mit.edu/courses/6-012-microelectronic-devices-and-circuits-fall-2005/4ed156440a15fea4df8f1e01eb310b_bb_cadence_calc.pdf

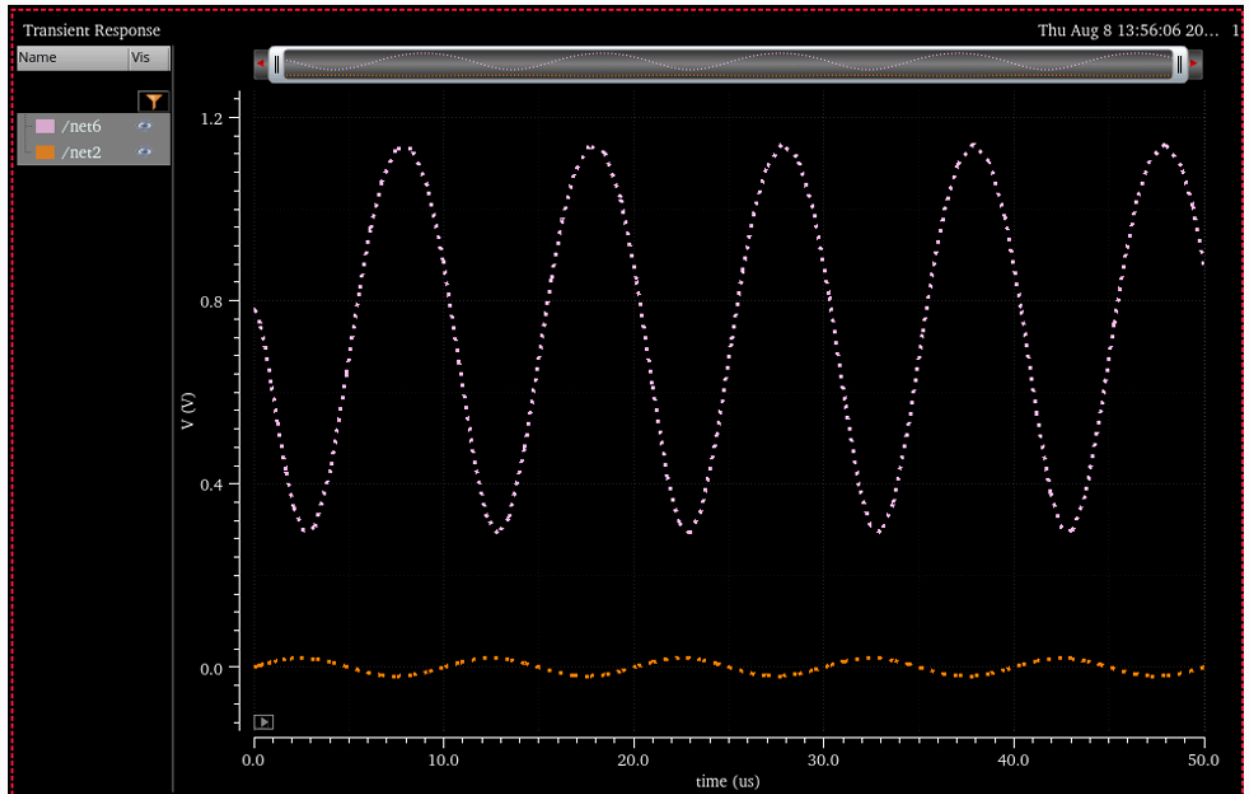
- dB20 is correct



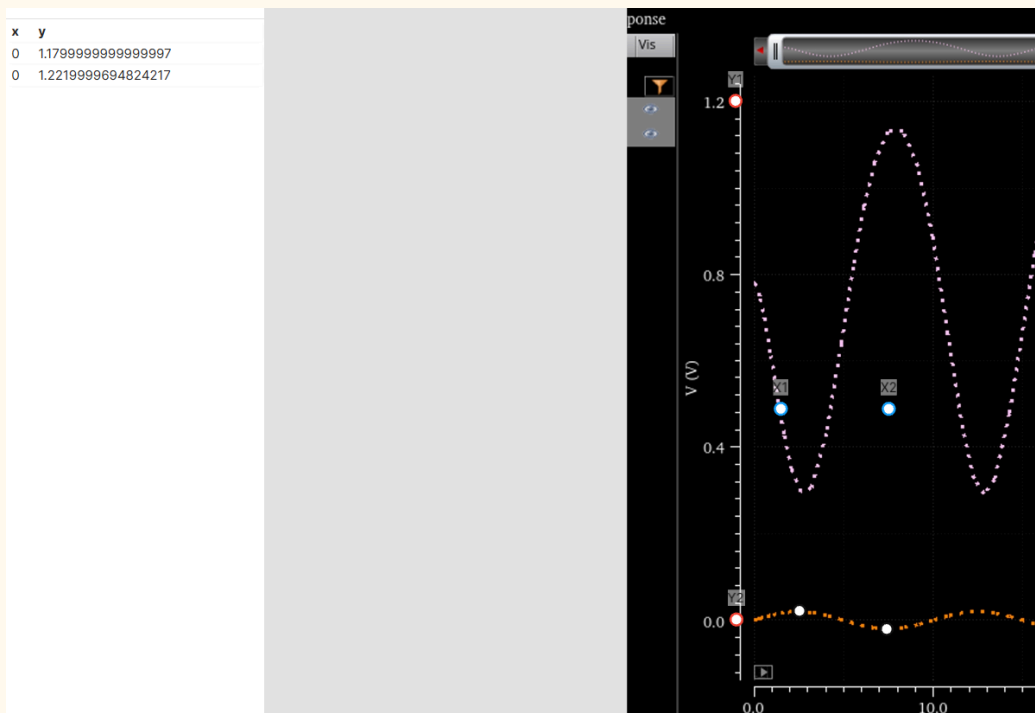
- wtf??

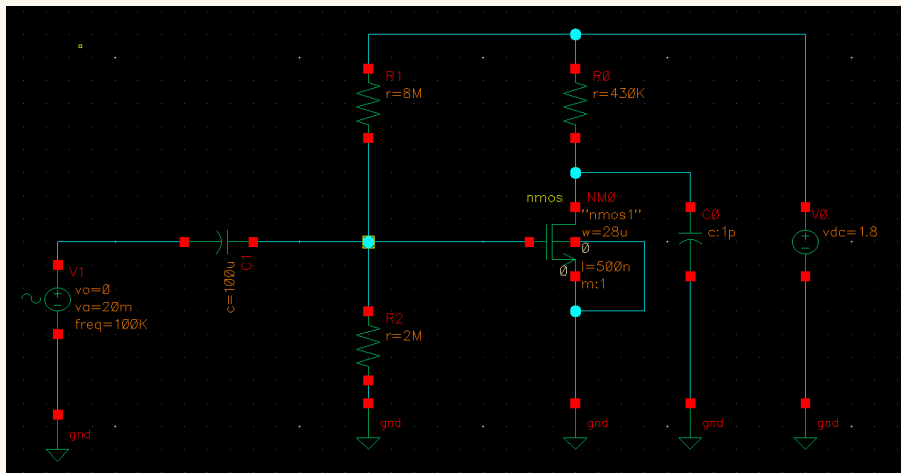
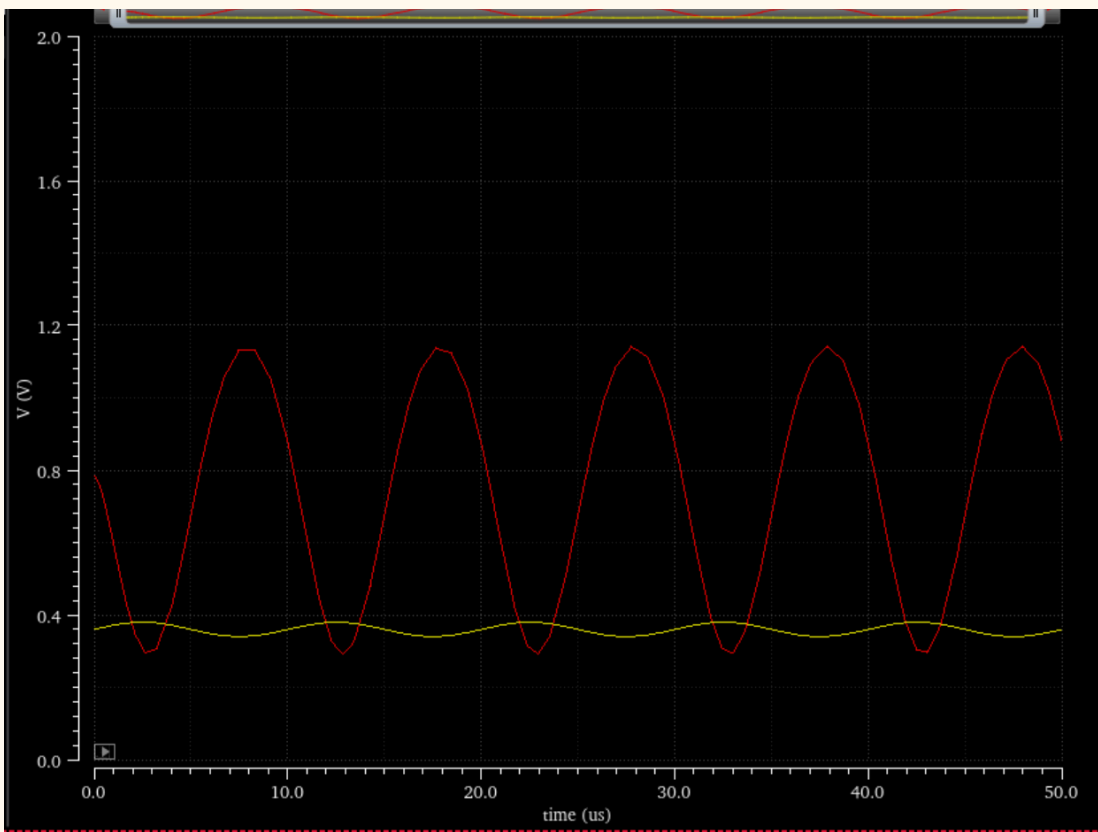
- how is this clipping with 2.7 dB ?????
- should not be this

Transient Result



- the pdf never clip also
- bruh their input is like $0.04 \text{ mV} = 40 \text{ mVpp}$





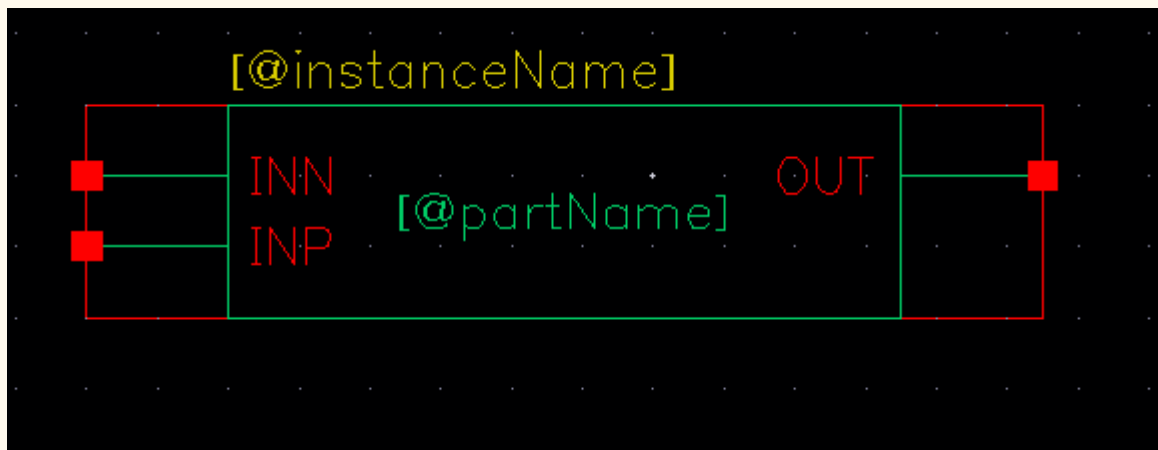
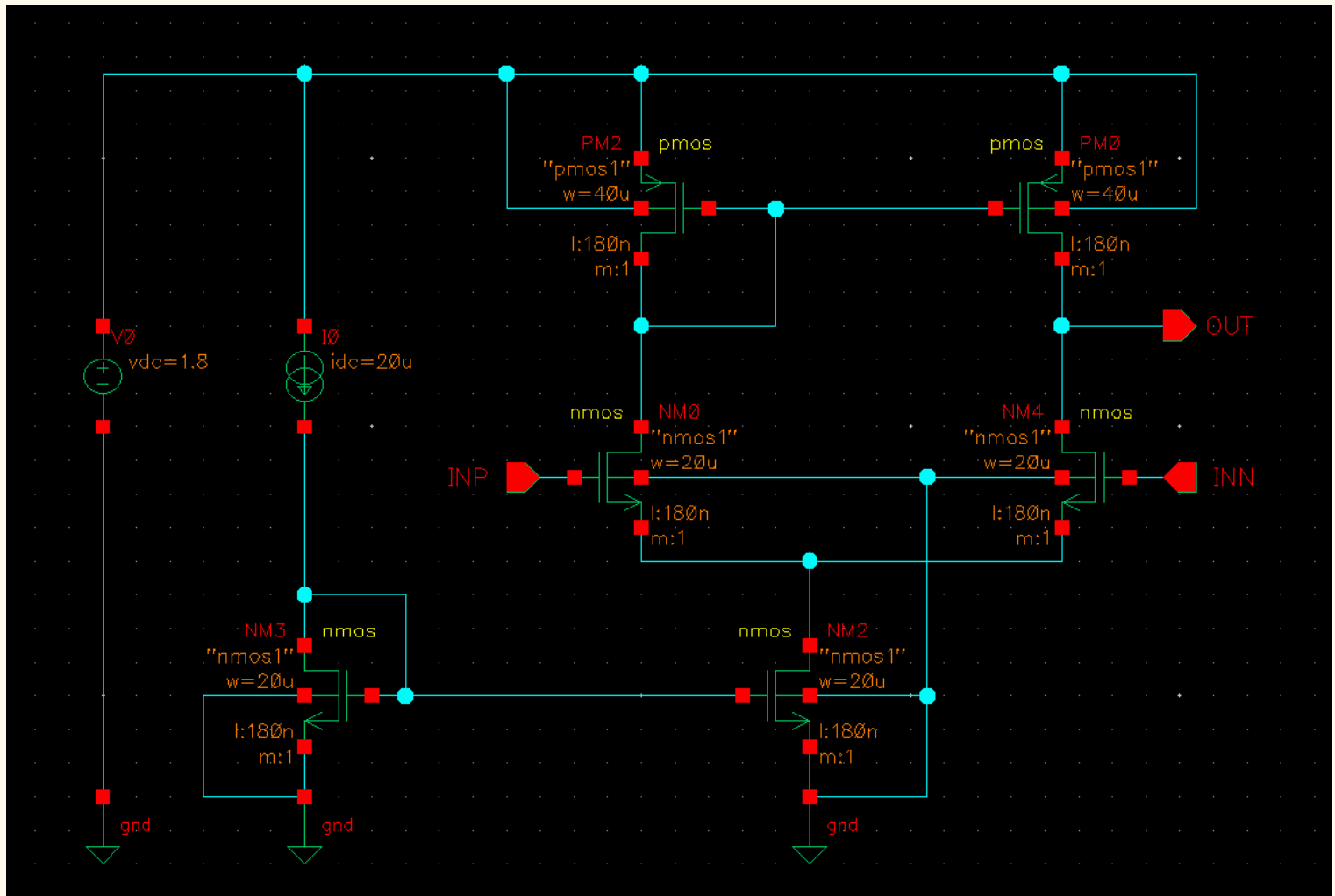
- seems to be right
- $10^{(26.8/20)} * 0.04V_{pp} = 0.875 V_{pp}$

● ah yes

Single Stage Amplifier

- Create the schematic of single stage amplifier similar to Lecture 3
- gpd180: nmos, analogLib: vdc, vsin, gnd, res, cap
- For vsin (V2 in the schematic above), set the AC magnitude to V, Offset voltage to 0V, Amplitude to 20mV and Frequency to 100kHz.

Part 2 Op Amp



https://community.cadence.com/cadence_technology_forums/f/custom-ic-design/25173/updating-the-symbol-pins-w-hile-doing-schematic-pin-change#:~:text=It%20doesn%27t%20happen%20automatically,that%20to%20you%20to%20decide.

- symbol will not get updated automatically if the cell view changes

Stability Simulation

- Refer to earlier slide for model library setup and choose stability analysis
- Choose “stb” under “Analysis”
- Select “Frequency” under “Sweep Variable”
- Under “Sweep Range”, set “Start” to “10k” and “Stop” to “1G”
- For “Probe Instance/Terminal”, type “/IPRBO” (this is the iprobe instance in the schematic, you can also click “Select” and directly choose from schematic)
- The good thing of using iprobe is to enable loop gain analysis without actually breaking the loop
- Run simulation by selecting **Simulation-> Netlist and Run**

Choosing Analyses -- ADE Explorer@lxb17.cadcam.nus.edu.sg

Analysis

☐ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☒ stb	☐ pz	☐ sp	☐ envlp
☐ pss	☐ pac	☐ pstb	☐ pnoise
☐ pxf	☐ psp	☐ qpss	☐ qpac
☐ qpnoise	☐ qpxf	☐ qpsp	☐ hb
☐ hbac	☐ hbstb	☐ hbnoise	☐ hbasp
☐ hbxf			

Stability Analysis

Sweep Variable

☒ Frequency

☐ Design Variable

☐ Temperature

☐ Component Parameter

☐ Model Parameter

☐ None

Sweep Range

☒ Start-Stop

Start: 10k Stop: 1G

☐ Center-Span

Sweep Type

Automatic

Add Specific Points ☐

Mode Type

☒ single-ended

☐ common

☐ differential

☒ Probe Instance

/IPRBO

☐ Probe Terminal

Local Ground Name

Enabled ☒

Options...

OK Cancel Defaults Apply Help

